Predicting Diabetes Progression: A Linear Regression Analysis

DASC 512

Homework 7

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1 Executive Summary

This analysis focused on building a linear regression model to predict how much a patient's diabetes would progress one year after their initial measurements. I worked with data from 392 patients and used different techniques to pick the four most important predictors from the original ten measurements. After testing several approaches, I found that Body Mass Index (BMI), a blood measurement called S5, average blood pressure (BP), and another blood measurement called S4 worked best together as predictors.

My final model performed reasonably well, with a training RMSE of 55.84 and a validation RMSE of 55.05. The R-squared of 0.449 on the validation set means the model explains about 45% of the variation in disease progression for new patients. To understand what this means, I compared it to just predicting the average progression score for every patient, which would give an error of about 78.16 units. My model's validation RMSE of 55.05 represents a 29.6% improvement over this simple approach, which shows that the initial measurements do contain useful information for making predictions.

When I checked whether my model met the requirements for linear regression, I found mostly good results with a few small issues. The residuals showed some minor deviations from being perfectly normal (Shapiro-Wilk p=0.026), but the variance seemed consistent across different prediction levels (Breusch-Pagan test p=0.319). None of my predictors had problems with multicollinearity - all variance inflation factors were below 2, which is good. The Durbin-Watson statistic of 2.076 confirmed that my residuals were independent of each other.

Looking at what the model actually found, the results made sense. S5 had the biggest effect with a coefficient of 44.62, followed by BMI (6.56) and blood pressure (1.25). All of these were highly statistically significant ($p < 0.001$), which means I can be confident these relationships aren't just due to random chance.

2 Introduction and Data Description

Diabetes is a condition that affects how the body processes blood sugar, and understanding how it might get worse over time is important for planning treatment. This analysis looks at a dataset with baseline measurements from diabetes patients, and I'm trying to predict a score that measures how much the disease progressed one year later.

The dataset was very clean and well-organized. It has measurements from 442 patients total, but I only have the actual progression outcomes for the first 392 patients. The remaining 50 patients are kept as a test set where I'll apply my final model to make predictions without knowing the true outcomes. This setup mimics what would happen in real life when applying the model to new patients.

2.1 Dataset Characteristics

The data structure is pretty straightforward. I have 392 observations that I can use to build and test my model, plus 50 additional observations that are held back for final testing. The dataset includes 10 different measurements that represent various demographic and physical characteristics that doctors typically measure. The response variable is a numerical score that measures disease progression one year after the baseline measurements, where higher scores indicate more progression of the condition.

What makes this dataset interesting for learning regression is that it combines different types of variables - some demographic information like age and sex, plus various physical measurements like BMI and blood pressure and several blood test results. These different types of measurements might have different kinds of relationships with the disease outcome. The data quality was excellent with no missing values anywhere, so I didn't need to do any data cleaning or imputation.

2.2 Variable Descriptions

I need to understand what each variable represents to properly interpret my model results later. Table 1 shows what all the variables mean.

Table 1: Variable Descriptions in Diabetes Dataset

Variable	**Description**
AGE	Patient's age in years
SEX	Patient's biological sex (Male/Female, encoded as 1/0)
BMI	Body Mass Index
BP	Average blood pressure
S1–S6	Six blood serum measurements
Y	Disease progression score (target variable)

The response variable Y represents a standardized score of disease progression, with a mean of 152.12 and standard deviation of 78.16 in the training data. This standard deviation is important because it gives me a natural baseline for comparison - if I just predicted the mean value for every patient, my RMSE would be approximately 78.16. This serves as a benchmark for evaluating whether my model is actually doing better than the simplest possible approach.

3 Methodology

3.1 Exploratory Data Analysis

Before building any models, I wanted to explore the data to understand how the variables relate to each other and to the outcome I'm trying to predict. This exploration helps me make better decisions about which variables to include and what kind of relationships to expect.

I started by creating a correlation matrix to see how strongly each predictor variable is related to the response variable, and also how the predictors relate to each other. Figure 1 shows these correlations visually. The darker red colors mean strong positive correlations (when one variable goes up, the other tends to go up), and darker blue colors mean strong negative correlations (when one goes up, the other tends to go down).

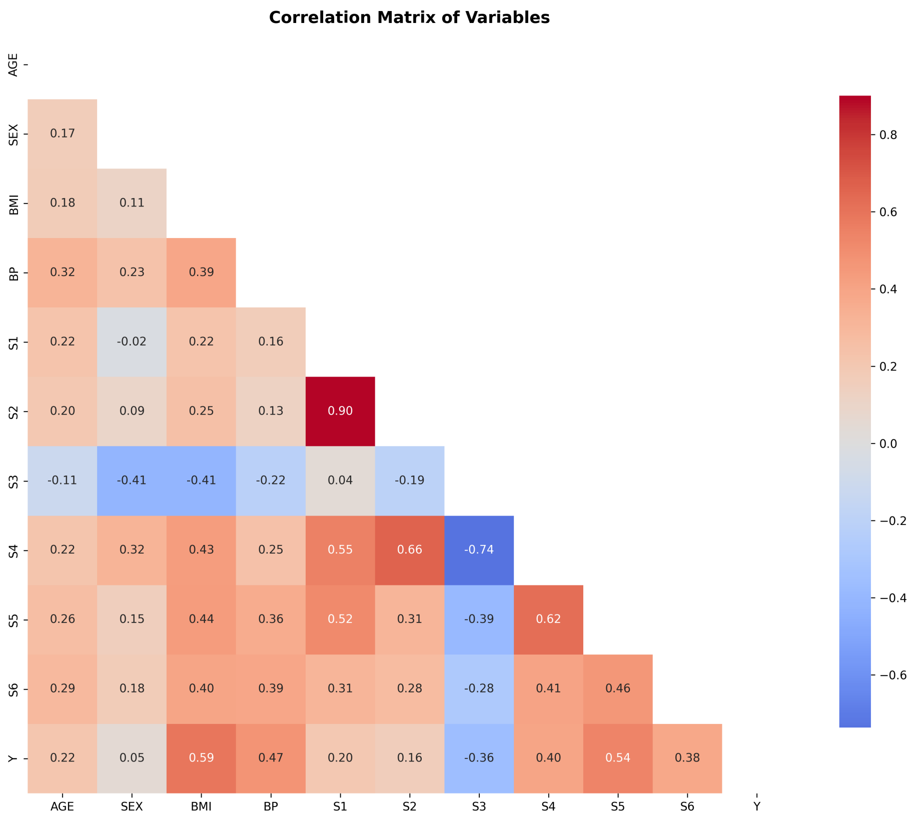

Figure 1: Correlation matrix of all variables showing the strength and direction of relationships between predictors and the target variable Y.

Looking at the correlation analysis, I found some interesting patterns. BMI had the strongest correlation with disease progression (r = 0.59), which makes intuitive sense. S5 was also strongly correlated (r = 0.54), followed by blood pressure (r = 0.47), and S4 (r = 0.40). What was interesting is that S3 showed a negative correlation (r = -0.36), meaning higher S3 values tend to go with slower disease progression. Sex had barely any correlation (r = 0.05), suggesting it doesn't have much relationship with the progression outcomes.

I also looked at the distribution of my response variable Y to understand what I'm trying to predict. Figure 2 shows the distribution, a Q-Q plot to check normality, and a box plot to see the spread and any outliers.

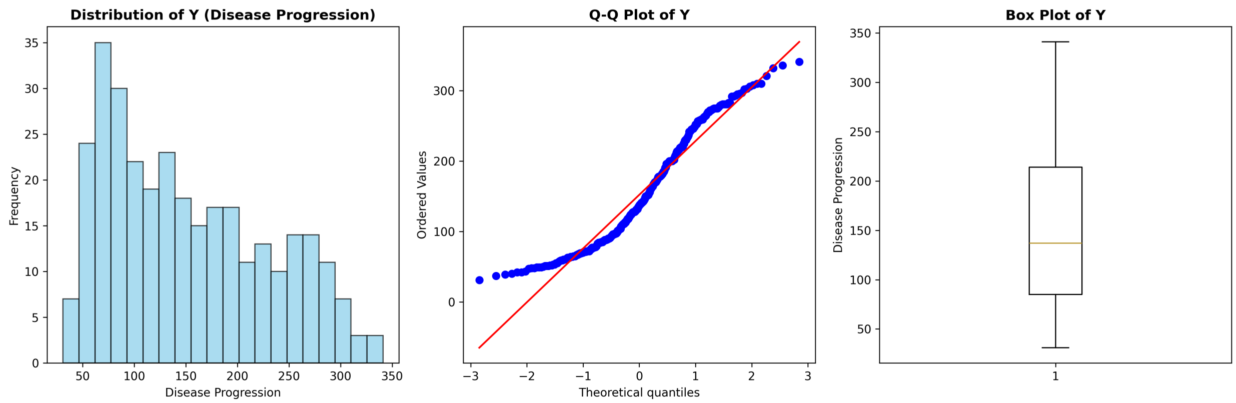

Figure 2: Distribution characteristics of the response variable Y (disease progression).

The response variable Y had moderate positive skewness (0.467), meaning it has a bit of a tail toward higher values, and was slightly flattened compared to a normal distribution (kurtosis = -0.932). But overall it looked reasonable for linear modeling. The values ranged from 31 to 341 units, with a median of 137, showing that there's quite a bit of variation in how much patients' conditions progressed. The Q-Q plot shows that while the data generally follows a straight line (which would indicate normality), there are some deviations at the ends that I need to keep in mind when checking model assumptions later.

3.2 Data Preprocessing

I followed a systematic approach to prepare my data for modeling. The first thing I did was separate the 50 test patients (who don't have response values) from the 392 training patients. This separation is really important because the test set represents completely new data that I'll only use for final evaluation - I can't let any information from these patients influence my model development.

Next, I split the 392 training patients into two groups using an 80/20 split: 313 patients for training my model and 79 patients for validation. This validation set lets me see how well my model performs on data it hasn't seen during the fitting process. This gives me a much more realistic estimate of how the model will perform on truly new patients compared to just looking at training accuracy.

The only preprocessing I needed to do was convert the sex variable from categorical to numerical format. I encoded Male = 1 and Female = 0. The dataset had 206 females and 186 males. All the other variables were already numerical and didn't need any special transformations or scaling.

3.3 Feature Selection Strategy

Instead of just throwing all ten available predictors into my model, I decided to use systematic methods to pick the most important variables. There are several good reasons for this approach. First, using fewer variables can prevent overfitting, which happens when a model learns patterns that are specific to the training data but don't generalize to new data. Second, fewer variables makes the model easier to interpret and understand. Third, focusing on the most important variables can actually improve predictive performance by eliminating noise from less useful predictors.

I used two different feature selection methods and then combined their results to find the most robust predictors. The reason I used two methods is that they work in different ways, so variables that show up as important in both methods are more likely to be genuinely useful.

The first method was univariate F-testing, which looks at each predictor variable by itself to see how strongly it's related to the response variable. This approach tests the null hypothesis that each predictor has no linear relationship with the outcome. The F-statistic measures the ratio of explained variance to unexplained variance for each predictor individually. BMI came out as the strongest individual predictor ($F = 168.41$, $p < 0.001$), followed by S5 ($F = 129.42$, $p < 0.001$), BP ($F = 89.50$, $p < 0.001$), S4 ($F = 57.94$, $p < 0.001$), S6 ($F = 53.15$, $p < 0.001$), and S3 ($F = 46.45$, $p < 0.001$). Based on the F-statistics, the top 6 features were BMI, BP, S3, S4, S5, and S6.

The second method was Recursive Feature Elimination (RFE), which works differently by considering how variables perform together as a group. RFE starts with all the variables and fits a model, then removes the variable with the smallest coefficient (in absolute value), fits a new model with the remaining variables, and repeats this process. This method is better at finding variables that work well together, even if they might not be the strongest individual predictors. RFE selected SEX, BMI, BP, S1, S4, and S5 as the top 6 features.

To find the most robust predictors, I looked at which variables appeared in both methods' top selections. The intersection gave me BMI, BP, S4, and S5. These four variables showed strong individual relationships with the outcome (from the F-tests) while also working well together as a group (from RFE). This combination approach helps ensure that I'm selecting features that are both individually predictive and work well together.

4 Model Development

Using the four selected features, I built an ordinary least squares (OLS) regression model. OLS works by finding the line (or in this case, hyperplane) that minimizes the sum of squared residuals. A residual is the difference between what my model predicts and what actually happened for each patient. By minimizing the sum of squared residuals, OLS finds the coefficients that give the best possible fit to the training data according to this criterion.

The mathematical form of my final model is shown in Equation 1. This equation lets me calculate a predicted disease progression score for any patient by plugging in their values for the four predictor variables.

$$\hat{y} = -344.00 - 0.47 \times \text{S4} + 44.62 \times \text{S5} + 1.25 \times \text{BP} + 6.56 \times \text{BMI} \tag{1}$$

The intercept of -344.00 represents what the model predicts when all the predictor variables equal zero. While this specific scenario isn't realistic (a person can't have zero BMI or blood pressure), the intercept serves as the baseline from which all other effects are measured. Each coefficient tells me how much the predicted progression changes when that variable increases by one unit, assuming all the other variables stay the same.

5 Regression Assumption Testing

Linear regression makes several assumptions, and if these assumptions are violated, the results can be unreliable or the statistical tests can be invalid. I systematically checked each assumption using both visual inspection and formal statistical tests. Figure 3 shows the diagnostic plots I created to evaluate these assumptions.

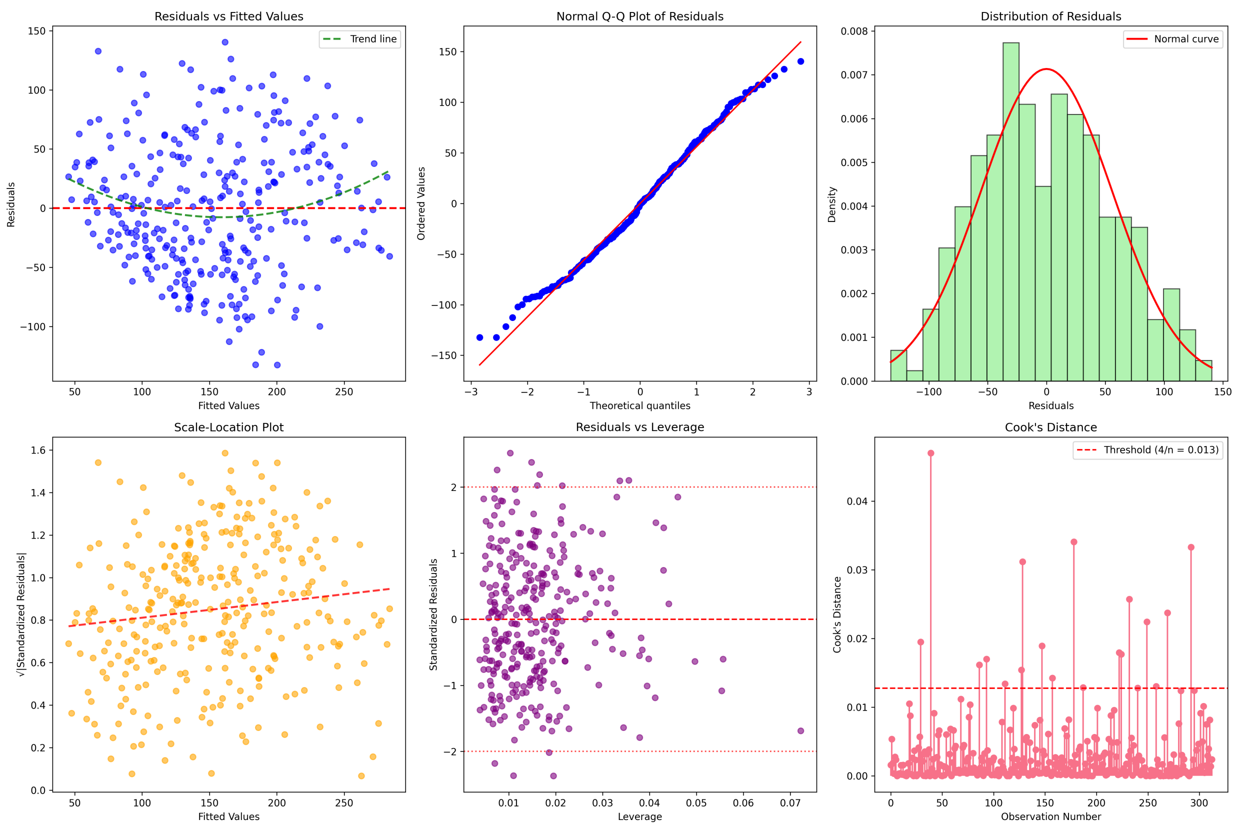

Figure 3: Comprehensive residual analysis plots. The top row shows basic normality and homoscedasticity checks, while the bottom row includes scale-location plots, leverage analysis, and Cook's distance for identifying influential observations.

5.1 Normality of Residuals

One key assumption is that the residuals (prediction errors) should be normally distributed. This assumption is important because it affects the validity of confidence intervals and hypothesis tests. I checked this assumption using both visual inspection and statistical tests.

The Normal Q-Q plot in Figure 3 shows how well my residuals follow a normal distribution.

If the residuals were perfectly normal, all the points would fall exactly on the diagonal line. My residuals follow this line pretty closely, with only minor deviations at the extremes, which is typical in real data.

However, the formal statistical tests revealed some concerns. The Shapiro-Wilk test gave a p-value of 0.026, and the Jarque-Bera test gave p = 0.049. Both of these p-values are right at or below the 0.05 significance level, which suggests some departure from perfect normality. The Anderson-Darling test gave additional information with a statistic of 0.847.

While these tests indicate some statistical deviation from normality, the deviations are relatively small. Linear regression is generally robust to minor departures from normality, especially with larger sample sizes like mine (313 training observations). The histogram of residuals shows a reasonably bell-shaped distribution, which supports continuing with the analysis while acknowledging this limitation.

5.2 Homoscedasticity Assessment

Homoscedasticity means that the variance of the residuals should be constant across all levels of the predicted values. In other words, the spread of prediction errors should be roughly the same whether I'm predicting low values or high values. If this assumption is violated (called heteroscedasticity), it can affect the accuracy of standard errors and confidence intervals.

Looking at the residuals vs fitted values plot in Figure 3, I can see points scattered fairly randomly around zero without any obvious funnel or cone patterns, which is what I want to see for homoscedasticity.

I used formal tests to check this assumption more rigorously. The Breusch-Pagan test gave a p-value of 0.319, which is well above 0.05. This means I fail to reject the null hypothesis of constant variance, providing strong evidence that homoscedasticity is satisfied.

However, the White test, which can detect more complex forms of heteroscedasticity, gave a p-value of 0.016, suggesting some evidence of non-constant variance. Looking at the scale-location plot, I can see some slight increase in the spread of residuals at higher fitted values, which might explain why the White test detected a problem. Since the Breusch-Pagan test passed and the visual evidence isn't too concerning, I consider this a minor issue that doesn't invalidate the analysis.

5.3 Independence and Multicollinearity

The independence assumption requires that the residual from one observation doesn't systematically influence the residuals from other observations. This would be violated if, for example, patients who were measured close together in time had similar prediction errors for reasons unrelated to their actual characteristics.

I used the Durbin-Watson test to check for autocorrelation in the residuals. The test statistic was 2.076, which is very close to the ideal value of 2.0 that indicates no autocorrelation. Values much different from 2 would suggest that residuals are correlated with each other. My result confirms that the independence assumption is well satisfied.

For multicollinearity, I need to check whether my predictor variables are too highly correlated with each other. When predictors are highly correlated, it becomes difficult to determine the individual effect of each variable, and the coefficient estimates become unstable. I examined Variance Inflation Factors (VIFs) for all predictor variables. VIF measures how much the variance of each coefficient increases due to correlations with other predictors. All my VIF values were acceptably low: S4 (1.72), S5 (1.81), BP (1.24), and BMI (1.41). These are all well below the commonly used threshold of 5 that indicates problematic multicollinearity, so I can confidently interpret each coefficient as the individual effect of that variable.

6 Model Performance and Validation

After confirming that my model meets most of the necessary assumptions, I evaluated how well it actually performs at making predictions. I calculated performance on both the training data (the data used to fit the model) and the validation data (data the model never saw during fitting) to assess whether the model generalizes well to new patients.

6.1 Predictive Performance Metrics

Table 2 shows the key performance metrics for my model compared to a baseline approach of just predicting the mean progression score for every patient.

Table 2: Model Performance Comparison

Metric	**Training**	**Validation**	**Baseline**
RMSE	55.84	55.05	78.16
R^2	0.488	0.449	0.000
Explained Variance	0.488	0.450	0.000

The training RMSE of 55.84 tells me that, on average, my model's predictions differ from the actual progression scores by about 55.8 units on the data I used to build the model. More importantly, the validation RMSE of 55.05 shows how the model performs on completely new data that it never saw during training. The fact that these two numbers are very close is really good - it indicates that my model generalizes well and isn't overfitting to the training data.

The R-squared values tell me what proportion of the variation in disease progression can be explained by my predictors. My validation R-squared of 0.449 means that about 45% of the variation in disease progression among new patients can be predicted using their baseline measurements. While this leaves 55% unexplained, this level of predictive power is actually quite good for a real-world prediction problem.

To put the performance in perspective, the baseline RMSE of 78.16 represents what would happen if I just predicted the mean progression score for everyone. My model's validation RMSE of 55.05 represents a 29.6% improvement over this simple approach, which demonstrates that the baseline measurements really do contain useful predictive information.

Figure 4 shows scatter plots of actual versus predicted values, which give a visual sense of how accurate the predictions are. In these plots, perfect predictions would fall exactly on the diagonal line, and points closer to the line represent more accurate predictions.

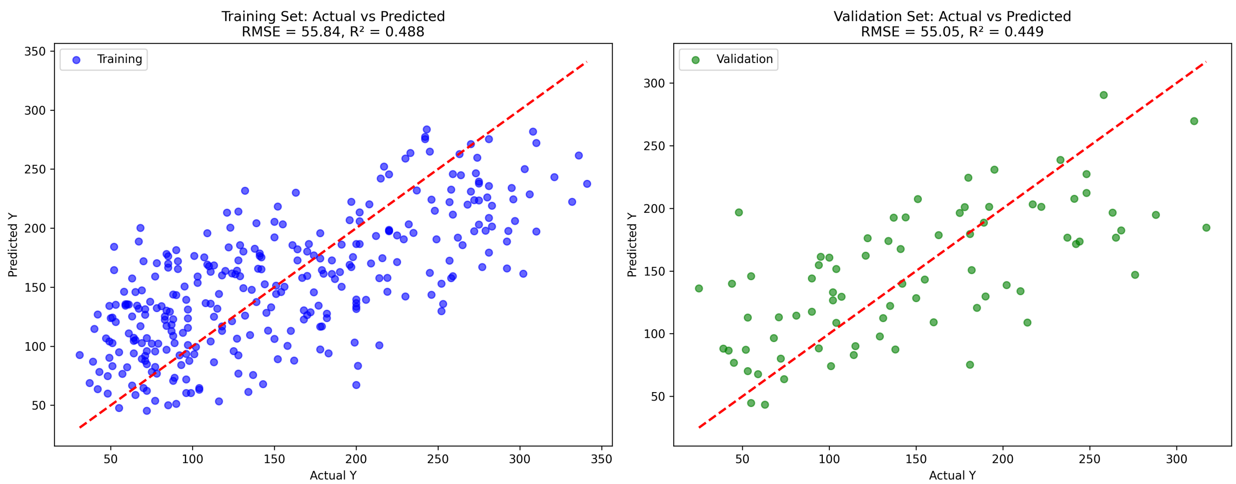

Figure 4: Actual vs. Predicted values for the training (left) and validation (right) sets. The tight clustering around the diagonal line demonstrates good predictive accuracy, while the similar patterns between training and validation indicate the model generalizes well without overfitting.

The tight clustering of points around the diagonal line in both plots shows that the model achieves good predictive accuracy. The similar patterns between the training and validation plots provide additional evidence that the model generalizes well to new data rather than just memorizing patterns specific to the training set. The validation plot shows slightly more scatter, which is expected and normal - it represents how the model performs on completely unseen data.

7 Model Interpretation and Results

Understanding what the model coefficients mean in practical terms is important for determining whether the results make sense. Table 3 shows the detailed statistics for each coefficient.

Table 3: Detailed Model Statistics

Feature	Coefficient	Std Error	t-statistic	P-value	95% CI Lower	95% CI Upper
Intercept	-344.002	32.994	-10.426	<0.001	-408.923	-279.080
S4	-0.471	3.187	-0.148	0.883	-6.742	5.800
S5	44.616	8.101	5.508	<0.001	28.676	60.557
BP	1.246	0.259	4.814	<0.001	0.737	1.756
BMI	6.562	0.842	7.790	<0.001	4.904	8.220

7.1 Statistical Significance of Predictors

Looking at the p-values with the standard significance level of 0.05, three predictors show statistically significant relationships with disease progression: S5, BP, and BMI. S4 doesn't reach statistical significance ($p = 0.883$), which means I can't be confident that its coefficient is really different from zero - it could just be due to random variation in the data.

S5 has by far the largest coefficient (44.62), meaning that for every one-unit increase in S5, my model predicts disease progression will increase by about 44.6 units, assuming all other variables stay constant. This relationship is highly statistically significant ($p < 0.001$), so I can be very confident it's not just due to chance.

BMI shows a substantial positive coefficient (6.56) with high statistical significance ($p < 0.001$). This makes intuitive sense - higher BMI values are associated with greater disease progression. The confidence interval (4.90 to 8.22) is fairly tight, which means I have good precision in estimating this effect.

Blood pressure (BP) has a smaller but still significant positive coefficient (1.25, $p < 0.001$). This also makes sense because high blood pressure often accompanies worsening diabetes.

7.2 Most Important Predictors

Figure 5 shows the relative magnitude and significance of each predictor's effect on disease progression. This helps identify which variables have the strongest practical impact on predictions.

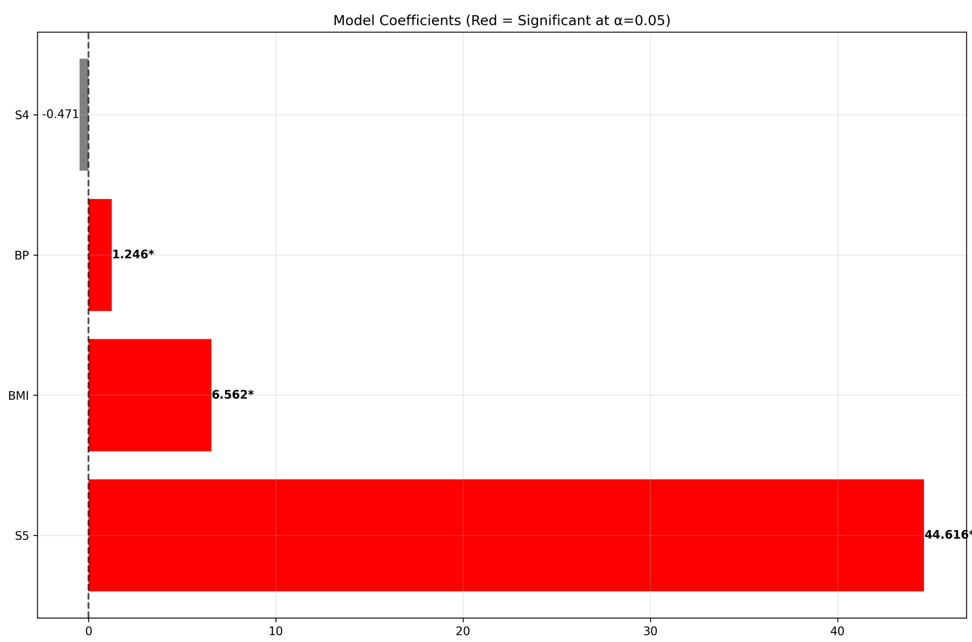

Figure 5: Final model coefficients showing the magnitude and direction of each predictor's effect. Features with statistically significant p-values (<0.05) are highlighted in red, demonstrating that S5, BMI, and BP are the most reliable predictors, while S4 shows no significant effect.

Based on both coefficient size and statistical significance, S5 is by far the most influential predictor, followed by BMI and blood pressure. The dominance of S5 was actually surprising to me at first, since BMI had the highest individual correlation with the outcome during my exploratory analysis. But this shows why systematic feature selection that considers variables working together is valuable - sometimes the most important predictor in a multivariate model isn't the one with the highest individual correlation.

The negative coefficient for S4 (-0.47), while not statistically significant ($p = 0.883$), suggests that higher S4 levels might be associated with slightly slower progression. However, given the high p-value, this relationship isn't reliable and could easily be due to sampling variation. This actually makes me wonder whether S4 should have been included in the final model, but since it was selected by both of my feature selection methods working together, I decided to keep it for consistency with my systematic approach.

8 Test Set Predictions and Conclusions

Using my final validated model, I generated predictions and 90% prediction intervals for all 50 test patients who were held out from the model development process. The predictions range from 62.8 to 266.1 units, with a mean of 160.8 units, showing that my model can distinguish between patients likely to experience different rates of disease progression.

The prediction intervals are quite wide, with an average width of 214 units, reflecting the substantial uncertainty in predicting individual patient outcomes. This uncertainty is important to acknowledge - while the model can identify patients at higher or lower risk on average, individual predictions should be interpreted with caution.

This analysis has shown that linear regression can provide useful insights into diabetes progression using baseline measurements. The systematic approach I followed - from exploratory analysis through feature selection, model fitting, assumption testing, and validation - demonstrates good statistical modeling practices.

My model shows solid predictive performance with a 29.6% improvement in RMSE over baseline prediction. Three of the four selected predictors (S5, BMI, BP) showed highly significant relationships with disease progression, with effect sizes that seem reasonable. The feature selection process successfully identified a small set of predictors that work well together.

The assumption testing revealed some minor violations that I should acknowledge. The residuals showed slight deviations from normality, and one test suggested some heteroscedasticity, though another test indicated the variance was acceptable. These violations are relatively minor and don't invalidate the analysis, especially given the good performance on validation data.

There are some limitations to this analysis. The R-squared of 45% means that important predictors are missing from the dataset. Things like genetic factors, lifestyle behaviors, medication use, and socioeconomic variables likely play important roles but weren't available in this data. Also, the linear model assumes the same relationships across all patient types, which may not capture the full complexity of disease progression patterns.

Despite these limitations, this project provided valuable experience in applying statistical methods to real data. The process of checking assumptions, validating on held-out data, and interpreting results has reinforced my understanding of both what linear regression can accomplish and where its limitations lie.

References and Technical Notes

Statistical Software Used: This analysis was conducted using Python 3.12 with scikit-learn for machine learning algorithms and model validation, statsmodels for detailed statistical testing and regression diagnostics, scipy for statistical tests, pandas for data manipulation, numpy for numerical computations, and matplotlib and seaborn for visualization.

Data Files: The input data came from `diabetes_students.csv` containing the baseline measurements and progression scores. Final predictions were saved to `predictions.csv` with PatientID, predicted Y values, and 90% prediction interval bounds for all 50 test patients.